

Disease Risk Prediction Using Decision Tree Algorithm

Integrating Wearable Sensor Technology & Machine Learning

Course: Artificial Intelligence and Expert Systems

Course Code: MLA0101

Topic: Decision Trees in Machine Learning

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Abstract

The Problem & Solution

Traditional disease diagnosis relies heavily on static, intermittent clinical visits, often missing early warning signs of chronic conditions.

This project proposes a machine learning-based predictive system utilizing continuous physiological data and clinical records for early detection.

Technology & Outcome

By leveraging the **Decision Tree Classification Algorithm**, the system intelligently maps medical attributes to risk outcomes.

The model evaluates input features (age, BMI, BP, sensor glucose/HR data) to output an interpretable, real-time prediction of **Low Risk** or **High Risk**.

Introduction

- 🏥 Healthcare Evolution: Predictive analytics is transforming healthcare by identifying high-risk patients before severe symptoms manifest.
- 🧠 Role of Machine Learning: ML algorithms analyze vast datasets and real-time physiological metrics significantly faster than manual methods.
- 📊 Why Decision Trees? In medicine, "black box" models are distrusted. Decision Trees mimic human logic, providing transparent, rule-based diagnostics.
- 🔧 Emerging Tech Integration: Wearable health sensors (smartwatches, IoT monitors) provide continuous vital sign streams, enabling dynamic AI predictions.



Problem Statement



Late Detection

Chronic diseases (e.g., cardiovascular issues) are frequently diagnosed at advanced stages due to infrequent clinical screening and reactive medical approaches.



Manual Limitations

Human interpretation of complex, multidimensional medical data is time-consuming, highly subjective, and prone to cognitive bias or physician fatigue.



No Continuous Data

Traditional systems fail to integrate real-time, continuous data from modern wearable sensors, completely missing sudden physiological anomalies.

"There is a critical need for an automated, highly interpretable machine learning system capable of integrating dynamic patient data to predict disease risk in real-time."

Project Objectives

- 🎯 **Develop an AI-Driven Model:** Design and implement a machine learning system using the Decision Tree algorithm for automated disease risk classification.
- 🎯 **Integrate Modern Sensor Data:** Effectively process physiological data streams (heart rate, BP) acquired from modern wearable IoT health sensors.
- 🎯 **Ensure Clinical Interpretability:** Generate clear, human-readable decision rules that medical practitioners can verify and trust over black-box models.
- 🎯 **Optimize Feature Selection:** Identify and isolate the most significant predictive features (BMI, glucose, age) to improve algorithmic efficiency.
- 🎯 **Evaluate Performance:** Rigorously assess model accuracy, precision, and recall against standard clinical validation datasets.
- 🎯 **Enable Proactive Care:** Output actionable "Low Risk" or "High Risk" predictions that directly aid in early medical intervention.

Analysis of the Existing System

Traditional Methodology

Currently relies on physical consultations, static laboratory blood tests, and manual observation of patient symptoms.

Doctors use generalized medical scoring systems (e.g., Framingham risk score) calculated manually or via basic spreadsheets based on point-in-time measurements.

System Limitations

- **Time-Consuming:** Processing lab results and manually calculating risk takes days.
- **High Error Margin:** Susceptible to human error and oversight of subtle correlations.
- **Static & Reactive:** Cannot utilize real-time continuous sensor data.
- **Scalability Issues:** Extremely difficult to track large populations simultaneously.

Proposed ML System

Automated ML Pipeline: We introduce an end-to-end framework that autonomously ingests patient demographics, clinical records, and wearable sensor telemetry.

Core Algorithm: The system utilizes a Decision Tree Classifier to recursively partition patient data based on the most critical health indicators.

The Interpretability Advantage: Unlike neural networks, this system provides *interpretable predictions*. Decision Trees use understandable rules (e.g., *If Glucose > 140 AND Age > 50*), allowing doctors to see exactly *why* a patient was flagged.



System Architecture Flow



Module 1: Data Collection & Preprocessing



Sensor Integration: Combines static records with dynamic data from optical sensors (e.g., smartwatch PPG sensors emitting green LEDs).



Data Ingestion: Importing CSVs, relational databases, and real-time IoT sensor APIs.



Handling Missing Values: Utilizing mean or median imputation techniques to ensure incomplete records don't crash the model.



Normalization: Scaling disparate sensor readings (e.g., heart rate vs. glucose) to a uniform standard for the algorithm.

Module 2: Feature Selection

Understanding the Process

Feature selection is the process of isolating the most relevant, influential variables from the dataset while dropping redundant or noisy data.

Primary Input Features Evaluated:

- Age, Gender, BMI
- Fasting Glucose Levels
- Systolic/Diastolic Blood Pressure
- Serum Cholesterol & Smoking Status

Why is it Important?

Improves Model Accuracy: Removes irrelevant data that could confuse the Decision Tree splitting logic.

Reduces Overfitting: Ensures the model learns actual risk patterns rather than memorizing training noise.

Optimizes Processing Speed: Absolutely vital for systems that process high-frequency, real-time data from wearable sensors, allowing for faster response times.

Module 3: Decision Tree Classification

Algorithmic Workflow

The tree splits data recursively into subsets based on feature values to maximize data homogeneity.

- **Root Node:** Most significant predictor.
- **Decision Nodes:** Conditional sub-questions.
- **Leaf Nodes:** Final predicted outcome.

```
Glucose > 140?  
├─ Yes → High Risk  
└─ No → Check BMI  
    ├─ BMI > 30 → High Risk  
    └─ Otherwise → Low Risk
```

Mathematical Splitting Concepts

Information Gain: Measures how much a specific feature reduces the overall entropy of the dataset.

Entropy: Measures the impurity or randomness in the data before a split occurs.

$$E(S) = - \sum_{i=1}^c p_i \log_2(p_i)$$

Module 4: Disease Risk Prediction



Executing the Final Prediction

Once trained, real-time patient data traverses the tree from the root to a leaf node. The specific path taken represents a verifiable clinical rule, ensuring complete diagnostic transparency.

Sample Output Interface:

```
=====
DISEASE RISK PREDICTION TOOL
=====
Patient ID: 49281 | Age: 45
BMI: 31 | Glucose: 145 mg/dL
-----
[ RUNNING DT CLASSIFIER... ]
>> RESULT: HIGH RISK (Conf: 89%)
>> ACTION: Recommend Cardiology Consult.
=====
```

Implementation & Technologies Used



Core Logic & ML Engine

Python serves as the primary programming language, utilizing the **Scikit-Learn** library for the rapid implementation of the `DecisionTreeClassifier`.



Data Processing Layer

Pandas and **NumPy** are heavily utilized for dataframe manipulation, cleaning missing values, and numerical transformations prior to model training.



Visualization & IDE

Developed using **Jupyter Notebook**. Data distributions, confusion matrices, and tree logic are visualized using **Matplotlib** and **Seaborn**.

Results and Performance Evaluation

The model is evaluated using Accuracy, Precision, Recall (Sensitivity - crucial in healthcare to find all actual high-risk patients), and the F1-Score.

Evaluation Metric	Traditional Rule-Based System	Proposed Decision Tree Model
Overall Accuracy	72.0%	89.5%
Recall (Sensitivity)	65.0%	91.2%
Processing Time	Hours / Days	< 1 Second
System Scalability	Low (Manual Effort)	High (Automated)

** Note: Values provided in this table are illustrative examples for presentation/prototype comparison purposes.*

Conclusion & Future Scope

Conclusion

The system successfully integrates machine learning with continuous medical sensor data. The Decision Tree model provides a highly accurate, automated, and interpretable tool that helps medical professionals predict disease risk swiftly and transparently.

Future Scope

- Integrate advanced continuous ECG patches and sweat-based biomarker sensors.
- Upgrade to ensemble methods (Random Forest / XGBoost) for even higher accuracy.
- Develop a mobile application for patients to monitor real-time predictive risk.

Thank You! Questions?